The EZ Diffusion Model

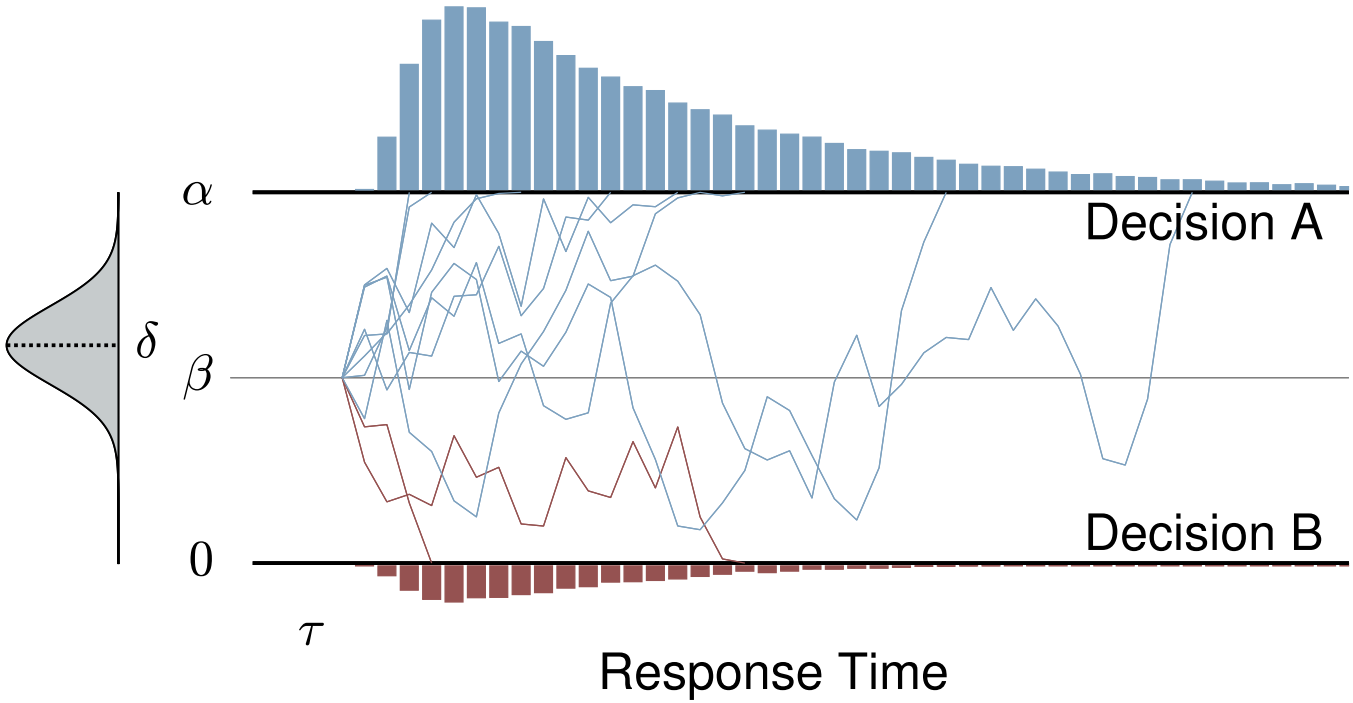
Michael D. Lee

Response Times

- Every behavior and every sequence of behaviors that people produce has a time course (Luce, 1986)
- A general framework for modeling response time is provided by sequential sampling models
  - a person collects information from a stimulus over time, and accumulates that information as evidence for making a response
  - once some critical level of evidence has been accumulated, the person makes a response
- The most popular and widely-used sequential sampling model of response times are drift diffusion models (Ratcliff, 1978; Ratcliff & McKoon, 2008)
  - usually applied to speeded tasks with discrete choices between two possible decisions
  - jointly models the decision and response time
  - based on what is known in statistics and physical as the Wiener diffusion process

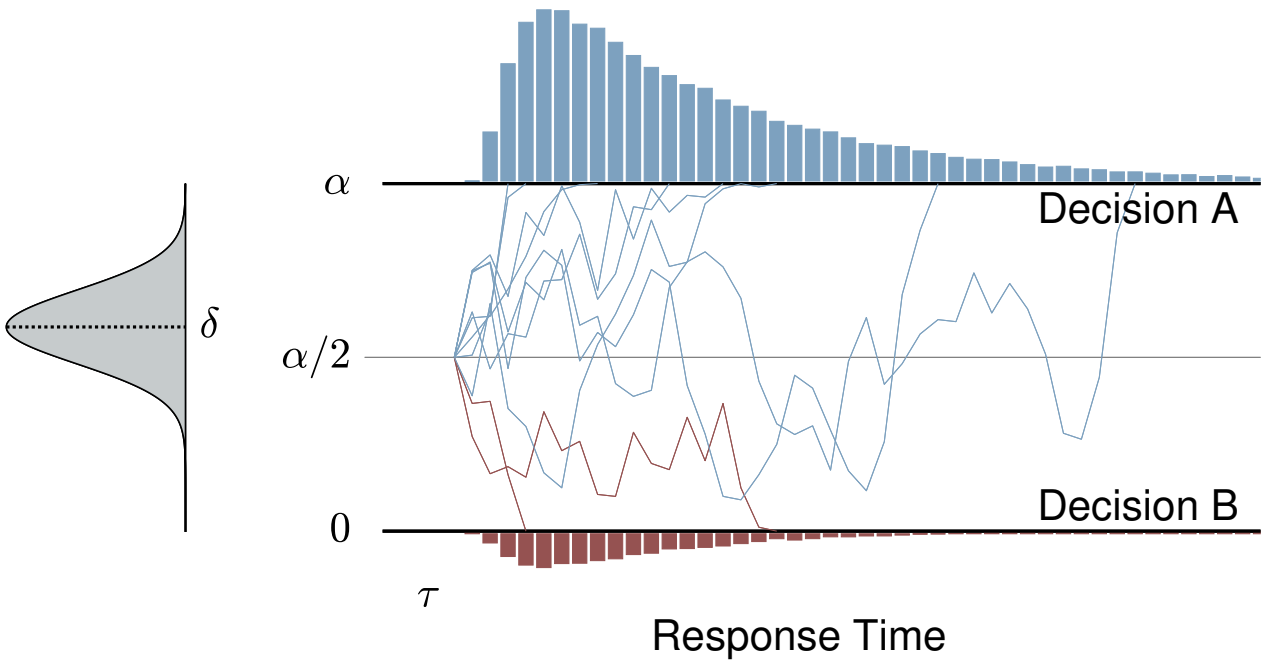
Wiener Diffusion

- The Wiener diffusion process has four parameters
  - evidence comes from a Gaussian distribution with mean drift rate δ
  - the samples of evidence are tallied towards two decision boundaries with boundary separation α
  - the evidence tallies all start at a bias β
  - the accumulation process starts after a time non-decision time has elapsed



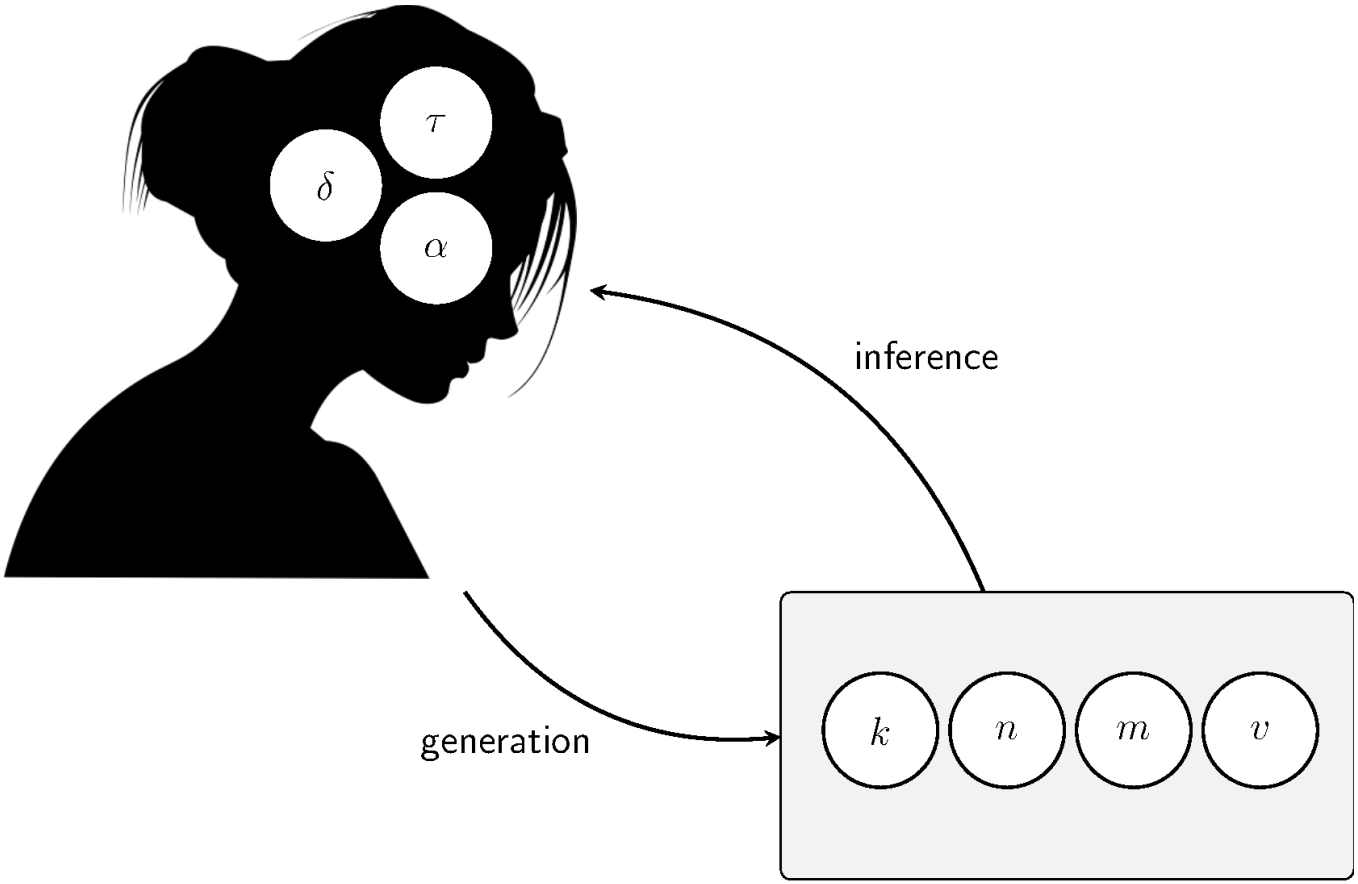
EZ Diffusion

- EZ diffusion is a simplified diffusion model that assumes that there is no bias (Wagenmakers et al., 2007; Chávez De la Peña & Vandekerckhove, 2025)
  - makes it suitable for implementation in JAGS and JASP
  - requires only summary decision and response time data



EZ Diffusion Model Structure

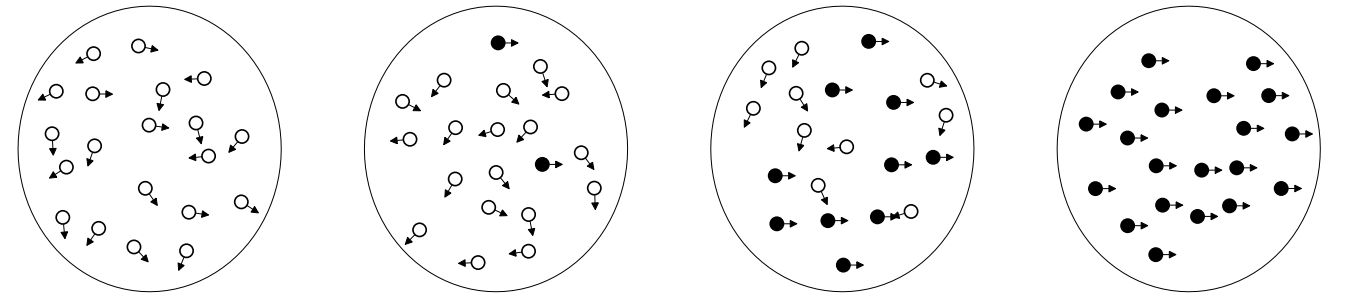
- The EZ diffusion model has the structure
  - observed number of decisions n, mean of response times for correct decisions m, and variance of response times for correct decisions v
  - drift rate δ, boundary separation α, and non-decision time τ parameters



Random Dot Motion

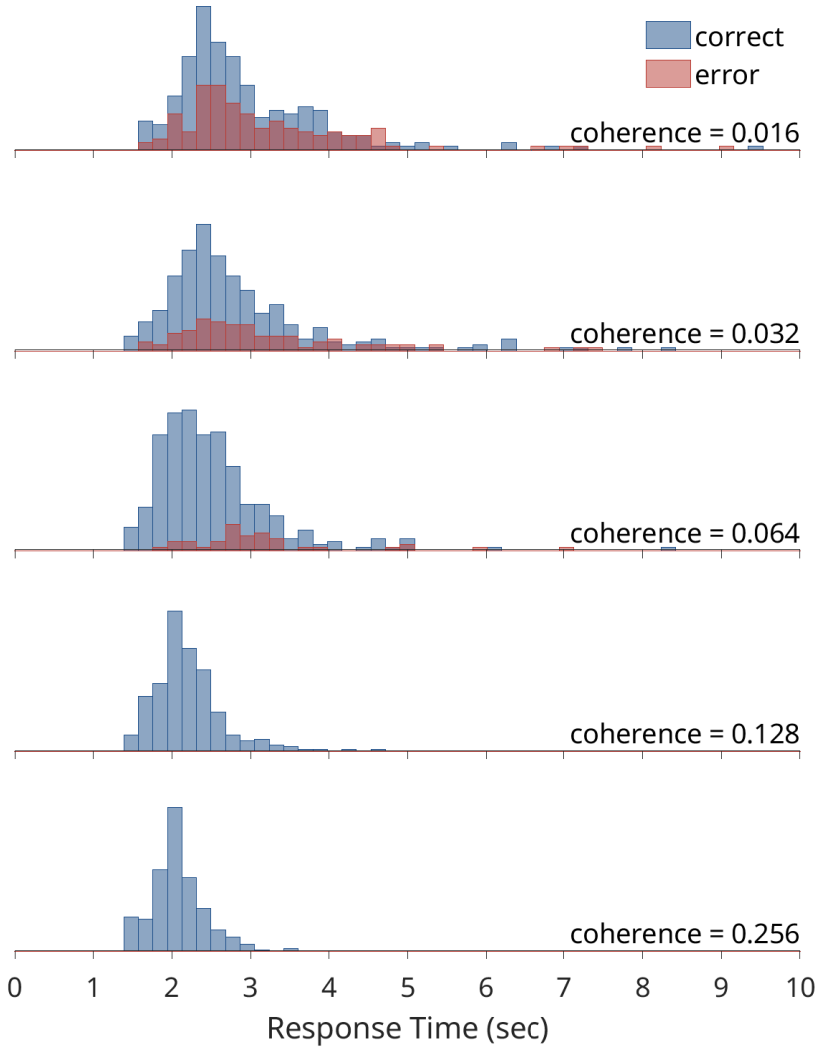
Random Dot Motion Tasks

- Random dot motion tasks are a common application for diffusion models
  - a stimulus contains a set of moving points
  - a subset move in the same direction while the others move in their own random directions
  - goal is to identify the direction of movement
  - different stimuli have different proportions of dots moving in the same direction, called the coherence of the stimuli
  - greater coherence generally makes detecting the direction of motion easier



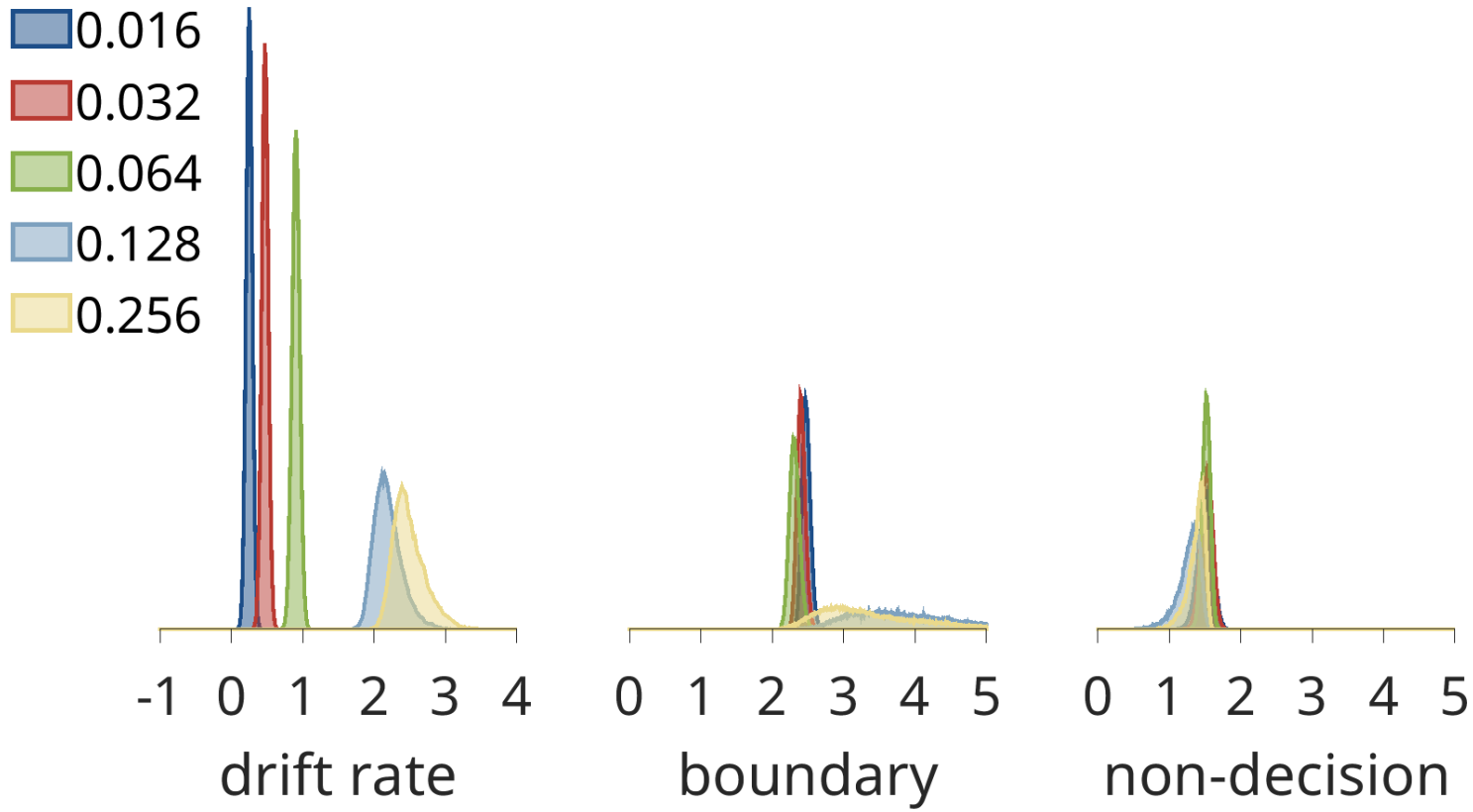
Hellmann, Zehetleitner, and Rausch (2023, Experiment 2) Task

- Participants decided whether the direction of motion was up or down for five coherence levels (1.6%, 3.2%, 6.4%, 12.8%, and 25.6%)
  - response time distributions for one participant are shown



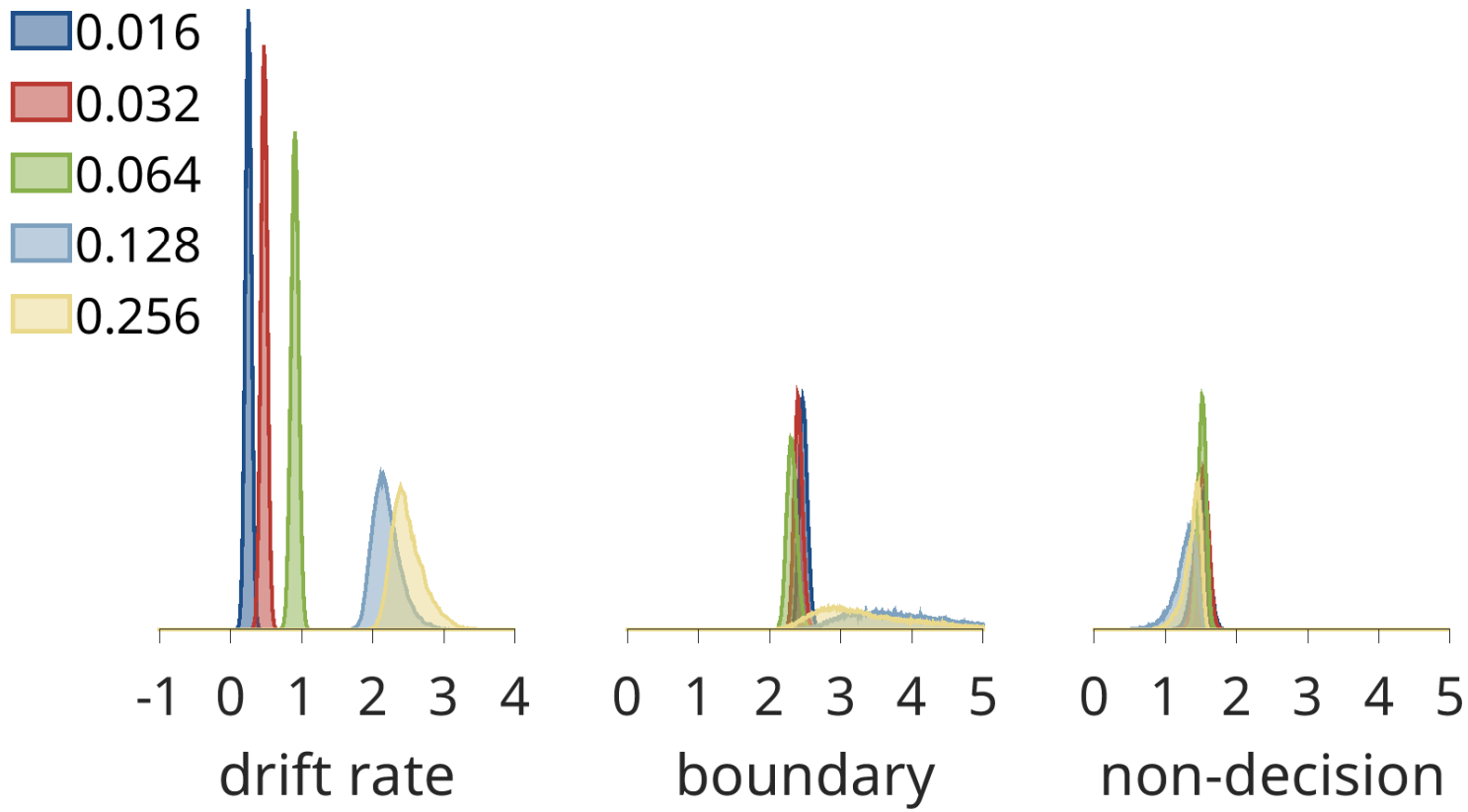
Results

- Marginal posteriors for the drift rate δ, boundary separation α, and non-decision time τ parameter for each coherence level are shown
  - drift rate generally increases with increased coherence
  - boundary separation and non-decision time do not clearly change



Results

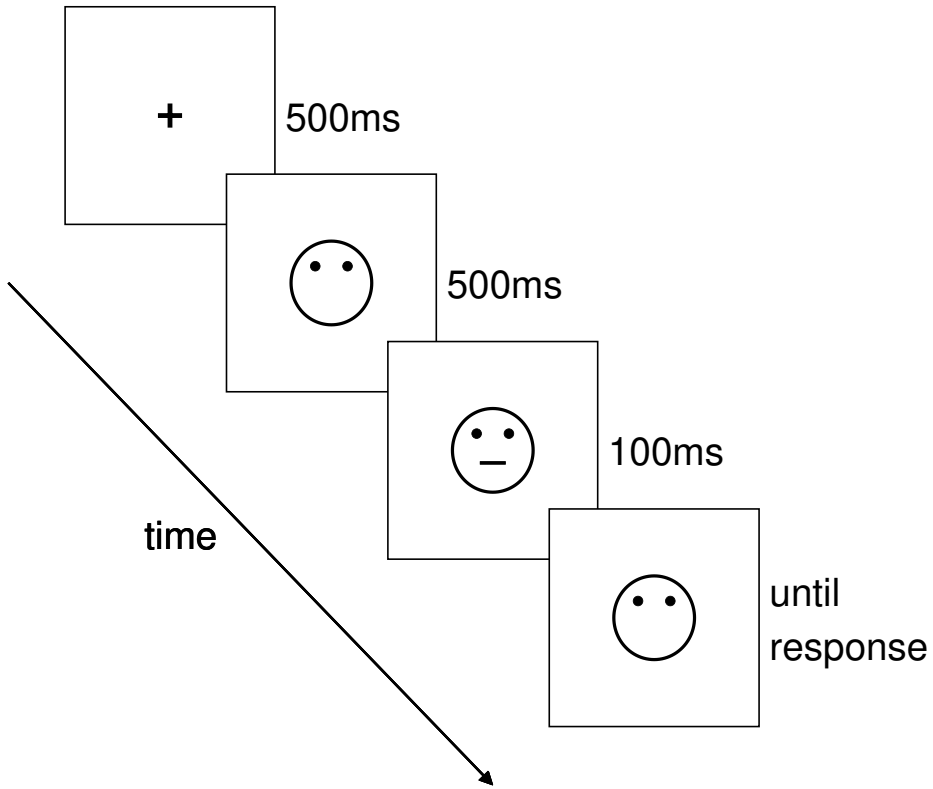
- The results are consistent with the conception that
  - drift rate measures the information in the stimulus
  - boundary separation measures the caution in the participant
  - non-decision time measures the encoding and motor response time of the participant



Speed-Accuracy Tradeoff

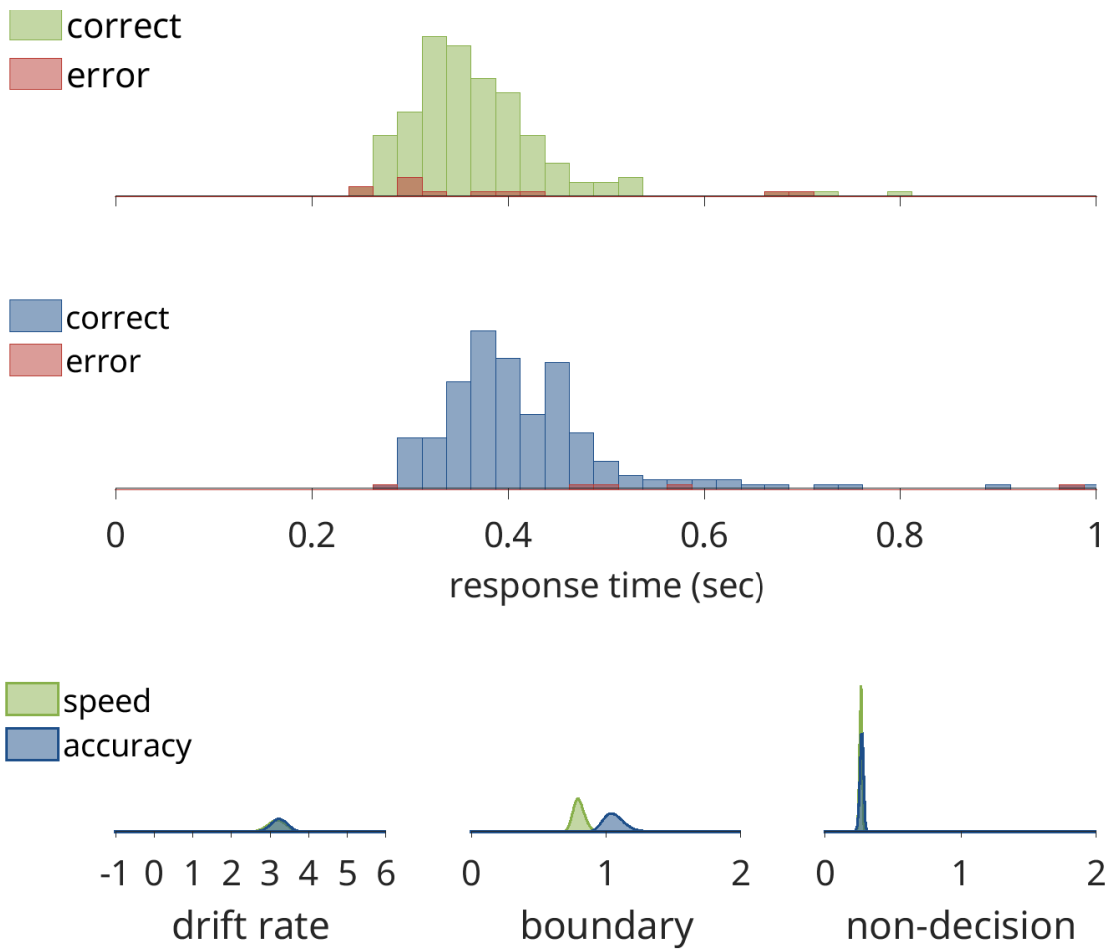
Perceptual Identification Task

- Grange (2022) studied differences in decision making between depressed and non-depressed participants in a perceptual identification task
  - participants decided whether the face had a long or short mouth
  - within-participant experimental manipulation alternated instructions emphasizing either making fast decisions or accurate decisions



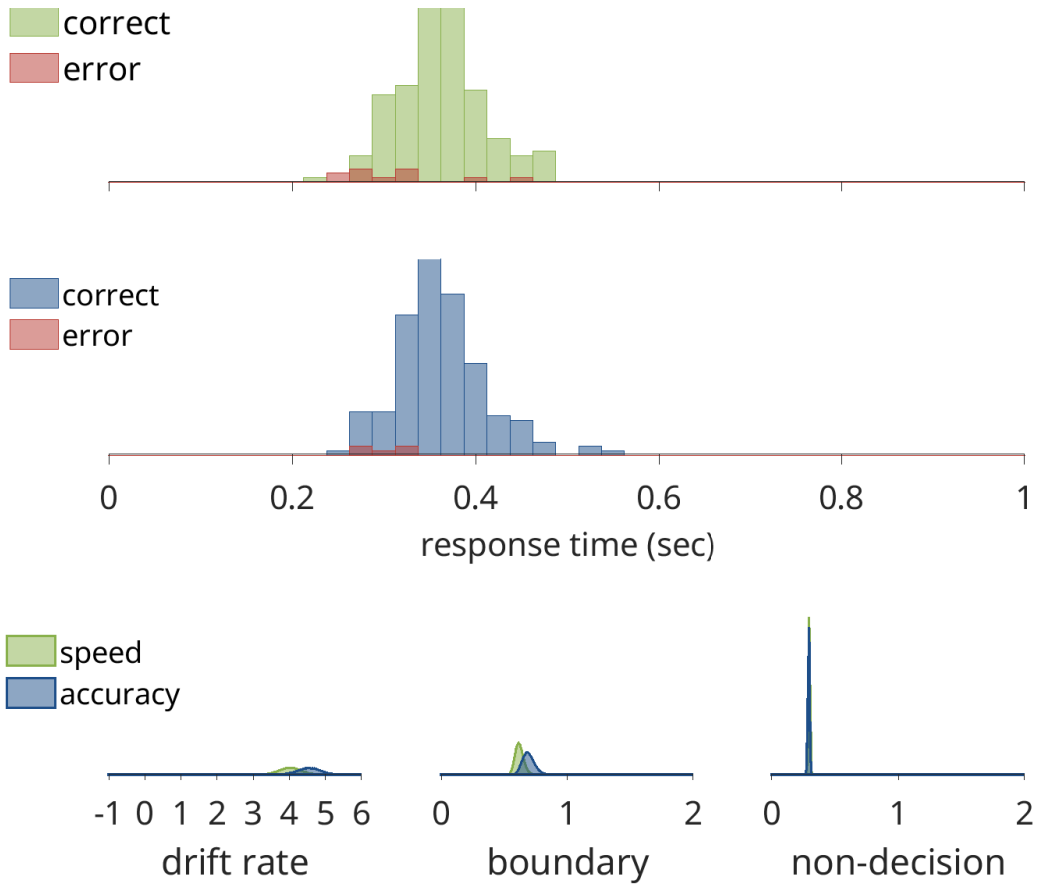
Participant A Data and Results

- The EZ diffusion model gives the theoretically expected results for Participant A
  - speed versus accuracy manipulation affects the boundary separation but not the drift rate or non-decision time



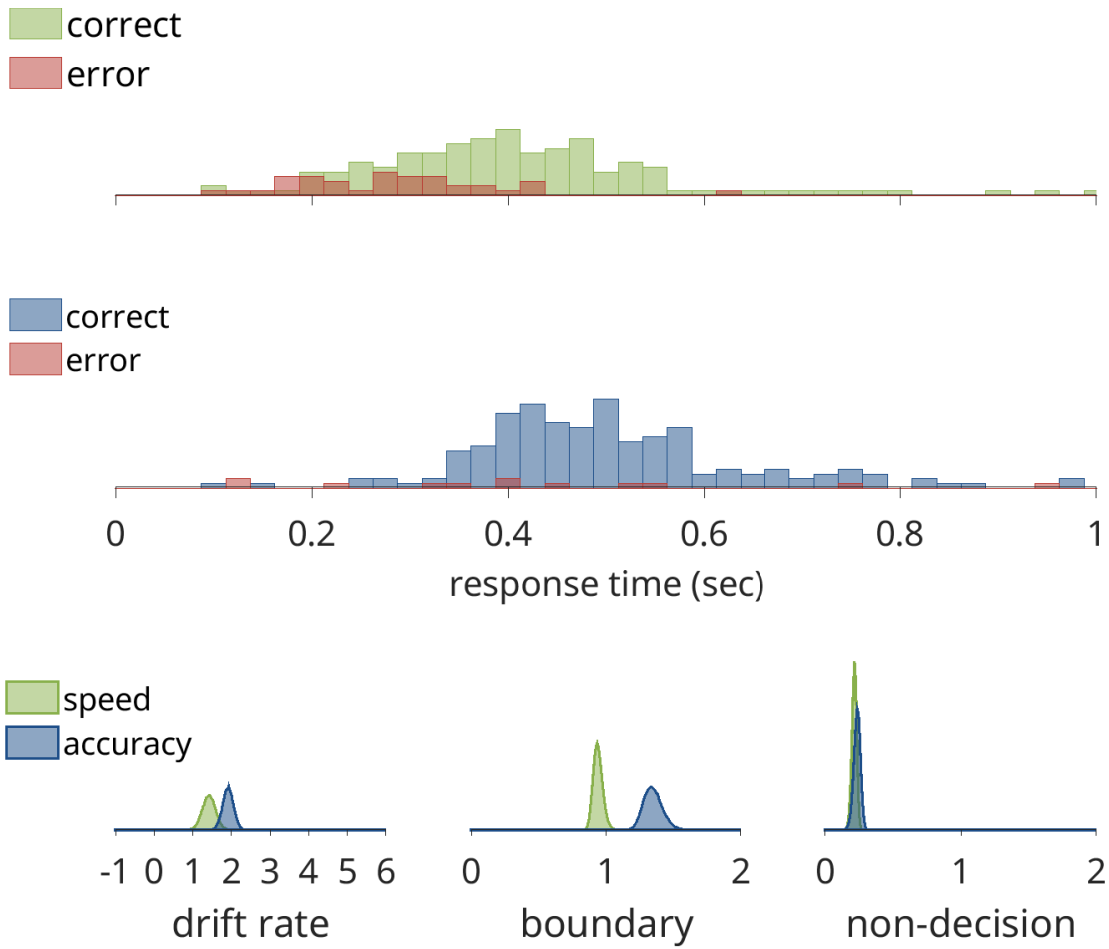
Participant B Data and Results

- The results are not as conceptually clean for Participant B
  - one interpretation is that the participant was able to extract more information but was more cautious in the accuracy condition
  - another interpretation is that the speed versus accuracy instructions did not impact the participant's behavior



Participant C Data and Results

- The results are also not conceptually clean for Participant C
  - one interpretation is that the larger number of errors in the speed condition is due to the participant being both less cautious and getting less information from the stimuli



References i

Chávez De la Peña, A. F., & Vandekerckhove, J. (2025). An EZ Bayesian hierarchical drift diffusion model for response time and accuracy. *PsyArXiv*.

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Ratcliff, R. (1978). A theory of memory retrieval. *Psychological Review*, *85*, 59–108. doi: 10.1037/0033-295x.85.2.59

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Ratcliff, R., & McKoon, G. (2008). The diffusion decision model: Theory and data for two-choice decision tasks. *Neural Computation*, *20*, 873–922. doi: 10.1162/neco.2008.12-06-

Wagenmakers, E.-J., van der Maas, H. J. L., & Grasman, R. P. P. P. (2007). An EZ-diffusion model for response time and accuracy. *Psychonomic Bulletin & Review*, *14*, 3–22.